

Assignment # 1

MAT 123

Dr. A H M MAHBUBUR RAHMAN

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Problems:

1. Consider that the defined function is

$$f(x) = \begin{cases} x^3 \cos(\frac{2}{x}) & x \neq 0 \\ 0 & x = 0 \end{cases}.$$

Show that $f(x)$ is continuous at $x = 0$.

2. Using the definition $\epsilon - \delta$, show that the above function is continuous at $x = 0$.

3. Find the differential coefficient of the following functions with respect to x :

$$y = (\sin(x))^{\cos(x)} + (\cos(x))^{\sin(x)}.$$

4. If $x = \sin(t), y = \sin(at)$, show that

$$(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + a^2 y = 0.$$

5. Find all n^{th} derivative of $\frac{1}{1-5x+6x^2}$.

6. i) Use the Mean-Value Theorem to show that if f is differentiable on an interval, and if $|f'(x)| \leq M$ for all values of x in the interval, then $|f(x) - f(y)| \leq M|x - y|$ for all values of x and y in the interval.

- ii) Use part (1) to prove that $|\sin(x) - \sin(y)| \leq |x - y|$ for all $x, y \in \mathbb{R}$.
7. Assume that f has a finite derivative in (a, b) and is continuous on $[a, b]$ with $a \leq f(x) \leq b$ for all x in $[a, b]$ and $|f'(x)| \leq \alpha < 1$ for all x in (a, b) . Prove that f has a unique fixed point in $[a, b]$.